QRAFT

Quantum-Resistant
Anonymous Frame Transactions

Use one unified private balance to send funds anonymously.

Built with ERC-8286 PQ wallets and EIP-8141 transactions.

Problems QRAFT addresses

Sender address exposure Submitting from a personal account reveals the submitting identity.	A shared proof-authorized account avoids using your personal account as the private-payment sender.
Quantum-vulnerable signatures Conventional account signatures lack post-quantum protection.	ERC-8286 PQ wallets use ML-DSA signing and ML-KEM receiving encryption.
IP exposure to RPC providers Direct RPC requests expose the user’s IP and request timing.	Anon RPC routes reads and deposit submissions through Nym.
Execution and gas complexity PQ authorization and gas sponsorship need a supported execution path.	Alternative mempool carries EIP-8141 authorization, sponsorship and execution frames.
Fragmented private funds Separate notes complicate tracking funds and paying exact amounts.	One unified private balance with automatic input selection and private change.

Private payments with one balance

Private ETH transfers

Send to a QRAFT receiving address within a shared pool. Deposit and withdrawal details remain public.

Automatic note management

Your device discovers incoming notes, totals unspent funds and selects payment inputs.

Exact amounts and private change

Send the requested amount. The remaining value returns to your private balance.

Passkey-controlled ERC-8286 PQ wallets

Unlock locally. ML-DSA signs transactions; ML-KEM encrypts incoming notes.

QRAFT and Tornado Cash Classic

Feature	Tornado Cash Classic	QRAFT
Payment flow	Fixed-denomination deposit and withdrawal	Exact private transfers and public withdrawals
Amounts and change	Withdraw a denomination, less applicable fees	Variable amounts; encrypted change notes
Note management	Deposit notes; optional encrypted Note Account	Automatic discovery, input selection and balance
Receiving funds	Withdrawal to a public recipient address	Encrypted payment to a QRAFT receiving address
Execution	Pool deposit and proof-based withdrawal calls	Sponsored EIP-8141 frame transactions

Scope: Classic only. Tornado Cash Nova already supports arbitrary amounts, shielded transfers and a shielded balance. This is a feature comparison, not a security ranking.

Sources: Classic deposit / withdrawal documentation · Nova documentation

Demo: a private ETH payment

Follow Alice and Bob through one payment.

01 · ALICE

Deposit 0.00004 ETH

Two deposits of 0.00002 ETH appear as one private balance.

02 · ALICE → BOB

Send 0.00003 ETH

Use Bob’s receiving address. Alice retains 0.00001 ETH privately.

03 · BOB

Receive 0.00003 ETH

Bob’s device discovers the encrypted payment and updates his balance.

04 · BOB

Withdraw 0.00001 ETH

Send to a public address. Bob retains 0.00002 ETH privately.

Switch to the app. Show confirmed results; label shortened waiting periods “Time compressed”.

How a private transfer works

01

Sender’s device

Selects notes, encrypts payment and change, generates a local proof.

→

02

Anon RPC + node

Browser submits EIP-8141 frames through Nym. Onchain contracts authorize and sponsor execution.

→

03

Private pool

Verifies proof and nullifiers, records commitments and encrypted notes.

→

04

Recipient’s device

Decrypts its notes, excludes spent notes and calculates the balance.

Deposit into the pool

Public ERC-8286 PQ account → ETH held by pool → private note.

Withdraw from the pool

Proof → pool sends ETH to public address.

Deposit and withdrawal amounts and addresses are public. Private-transfer amounts are hidden by the proof and encrypted notes.

Implemented and tested

Verified flow

93 tests pass. Local real-proof tests pass. The new relayerless live flow is undergoing verification.

Cryptography

PQ account signatures and receiving encryption. Current Groth16 proofs are conventional cryptography.

Network privacy

All payment submissions use Nym Anon RPC, without an application relayer. Network correlation remains possible; full Nym E2E testing is in progress.

Prototype status

Development proving setup. No independent security audit or production-security claim.

github.com/Shoydon/QRAFT · Implementation and test evidence